

S1 FUNCTIONAL REQUIREMENTS ENGINEERING

Smart Multi-Warehouse Stock Redistribution System — S1 Requirements & Engine Workbook

Common Team Information

Field	Student Entry
Team No.	2
Project Theme / Title	Smart Multi-Warehouse Inventory Redistribution System
Guide Name	KMMR
Date Prepared	04-Oct-2026

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SECTION A — DETAILED ENGINE DECOMPOSITION

1. What Students Must Establish Before Writing Requirements

Start from the approved S0 problem, stakeholders, pain points, scope and expected system behaviour. Identify the major technical responsibilities required to make the proposed system work. Group closely related responsibilities into engines that can be independently explained, owned, implemented and tested.

At this point, the engine list is a justified technical decomposition of the approved problem. After the SRS is prepared, students must return to the engine section and add the final requirement IDs supported by each engine. This keeps the decomposition before requirements while preserving traceability.

2. Engine Decomposition Rule

Step	Student Action	Evidence to Write
1. Problem Function	Identify a major system function or technical challenge from S0.	State the function/problem clearly.
2. Responsibility Group	Group functions that need the same technical mechanism, state or control.	Explain why the functions belong together.
3. Technical Responsibility	Convert the group into one precise engineering responsibility.	Write one responsibility statement beginning with “This engine is responsible for...”
4. Engine Boundary	Decide what the engine owns and what it must not own.	List in-scope and out-of-scope responsibilities.
5. Engine Definition	Name the engine based on the technical responsibility, not a screen/layer.	Give engine name, purpose, I/O, state, mechanism, failures and KPI.
6. Ownership	Assign one primary student owner.	Owner must defend the engine end-to-end.
7. Requirement Link	After SRS completion, map FR/NFR IDs to the engine.	Update the requirement mapping fields before Gate 1.

STUDENT CHECK: Avoid engine names such as “Frontend”, “Backend”, “Database”, “Login Page” or “Admin Screen” unless a distinct technical responsibility is demonstrated. The PPT example decomposes one integrated system into independently verifiable engines such as scheduling, routing, transaction management and monitoring.

4. Proposed Engine Summary — Before Requirements

Engine ID	Proposed Engine Name	Major course Covered	Technical Problem Owned	Primary Responsibility	Primary Student Owner
E1	Demand & Inventory Risk Engine	DBMS+DAA	Static stock thresholds alone cannot capture DOI, demand velocity, EOQ, reservations and changing source/destination demand.	Maintain inventory state, calculate replenishment/redistribution signals and identify stock-risk conditions.	Shrinidhi K
E2	Warehouse Ranking & Route Optimization Engine	DSA	Nearest-only selection can choose an unsafe or expensive source; a long-distance mother warehouse may be needed only as a fallback.	Filter feasible sources, rank them using weighted multi-factor scoring, support multi-source fulfillment and compute routes.	Amruta J
E3	Transfer Transaction & Inventory	DBMS-SE	Concurrent requests, retries and partial receipts can cause	Reserve stock transactionally, enforce transfer state transitions,	Kiran

Engine ID	Proposed Engine Name	Major course Covered	Technical Problem Owned	Primary Responsibility	Primary Student Owner
	Consistency Engine		double allocation or inconsistent stock.	update balances and maintain audit records.	
E4	Operations Workflow & Monitoring Engine	WEBT	Operational decisions need role-based actions, approvals, logistics assignment, status tracking, dashboards and notifications.	Coordinate request-to-receipt workflow, role permissions, logistics status, dashboard views and alerts.	Rajat P

EVIDENCE 1 OF 7 — SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
Stage S1 — Functional Requirements Engineering | General, Theme-Independent

1. Functional Requirements (FR)

Functional baseline: 22 testable functional requirements covering inventory, demand, redistribution, routing, transactions, logistics, monitoring and auditability.

FR ID	Description	Priority / (Mandatory/Challenge)	Source Need / (from S0)
FR-01	The system shall authenticate authorized users and provide role-specific access to permitted functions.	Mandatory	Operational security and role-based workflow.
FR-02	The system shall maintain product, warehouse and inventory information, including available, reserved and safety-stock quantities.	Mandatory	Reliable distributed inventory visibility.
FR-03	The system shall monitor inventory levels across all participating warehouses.	Mandatory	Central visibility of stock position.
FR-04	The system shall calculate Economic Order Quantity (EOQ) for replenishment planning where the product has applicable demand/cost parameters.	Mandatory	Support systematic replenishment planning; intra-day redistribution uses current risk/DOI.
FR-05	The system shall analyze historical sales and demand trends to derive demand velocity for products.	Mandatory	Detect demand changes rather than relying only on static thresholds.
FR-06	The system shall classify product inventory status as healthy, low or critical using configurable stock thresholds and demand-aware indicators.	Mandatory	Early identification of service-risk conditions.
FR-07	The system shall automatically generate a stock redistribution request when Days of Inventory (DOI) falls below the configured trigger (baseline: DOI < 1), subject to validation of current stock and demand data.	Mandatory	Trigger redistribution before local availability is affected.
FR-08	The system shall compute a valid route between source and destination warehouses using a weighted warehouse graph and shortest-path algorithm where applicable.	Mandatory	Reduce transfer travel distance/time after a source candidate is selected.
FR-09	The system shall identify eligible source warehouses using transferable stock, safety-stock constraints, warehouse operational status and current source demand risk.	Mandatory	Never recommend a source that cannot safely spare inventory.

FR ID	Description	Priority / (Mandatory/Challenge)	Source Need / (from S0)
FR-10	The system shall rank eligible source warehouses using a configurable weighted scoring mechanism based on distance, transferable surplus, source risk, transfer cost and ETA.	Mandatory	Choose the best feasible source instead of nearest-only selection.
FR-11	The system shall calculate a recommended transfer quantity that satisfies destination need without reducing the source below its safety-stock boundary.	Mandatory	Prevent solving one shortage by creating another.
FR-12	The system shall allow an authorized warehouse/operations manager to review, approve or reject a transfer request.	Mandatory	Retain human operational control over redistribution.
FR-13	The system shall reserve approved source inventory transactionally before dispatch.	Mandatory	Prevent double allocation of the same stock.
FR-14	The system shall prevent over-allocation when multiple transfer requests for the same stock are processed concurrently.	Mandatory	Handle concurrency safely at the DBMS layer.
FR-15	The system shall automatically assign an available logistics employee/resource to an approved transfer.	Mandatory	Reduce manual coordination delay after approval.
FR-16	The system shall track transfer status through Assigned → Picked Up → In Transit → Delivered → Received with timestamps.	Mandatory	Provide end-to-end operational visibility.
FR-17	The system shall update source and destination inventory after transfer receipt is confirmed and verified.	Mandatory	Keep inventory state synchronized with physical movement.
FR-18	The system shall support partial/multi-source fulfillment when one eligible warehouse cannot safely satisfy the complete requested quantity.	Mandatory	Use multiple safe sources rather than forcing an unsafe single-source transfer.
FR-19	The system shall support mother/regional warehouse replenishment as a fallback when suitable local stock is unavailable, while considering long-distance transfer cost, ETA and source safety.	Mandatory	Handle regional shortages without treating the mother warehouse as the default source.
FR-20	The system shall maintain transfer history and an audit log for approvals, reservations, assignments, status changes and inventory changes.	Mandatory	Provide traceability and accountability.
FR-21	The system shall provide role-based dashboards showing inventory levels, stock-risk states, requests, recommendations, transfers and operational status.	Mandatory	Give each role the information needed for its decisions.
FR-22	The system shall notify relevant users when a request is created, approved, rejected, assigned, delivered or completed.	Mandatory	Keep stakeholders synchronized with workflow state changes.

2. Non-Functional Requirements (NFR)

Non-functional baseline: 12 measurable quality requirements covering performance, reliability, security, integrity, concurrency, availability, scalability, maintainability, usability, auditability, recoverability and compatibility.

NFR ID	Description	Category	Measurable Target
NFR-01	Normal user operations and dashboard/API requests shall respond within an acceptable operational time.	Performance	>=95% of normal requests <= 2 seconds on the five-warehouse pilot dataset.
NFR-02	Inventory and transfer updates shall remain consistent even if an operation fails or the application/process is interrupted.	Reliability / DBMS	0 partial committed transfer states in recovery tests.
NFR-03	Only authenticated and authorized users shall access functions permitted for their role.	Security	100% of unauthorized access/approval tests rejected.

NFR ID	Description	Category	Measurable Target
NFR-04	The system shall prevent negative inventory and inconsistent source/destination quantities.	Data Integrity	0 negative balances and 0 unreconciled balance mismatches in validation tests.
NFR-05	Concurrent transfer requests shall be handled safely without double reservation or double allocation.	Concurrency / DBMS	100% of concurrency test cases preserve a single valid allocation.
NFR-06	The system shall remain available to authorized users during normal warehouse operating periods.	Availability	>=99% application availability during the pilot test window, excluding planned maintenance.
NFR-07	The system shall support adding warehouses, products, users and transactions without major architectural redesign.	Scalability	Demonstrate adding at least 5 more warehouses and 2x pilot transaction volume without schema redesign.
NFR-08	The system shall use modular components so inventory, ranking/routing, approval/logistics and reporting functions can be changed independently.	Maintainability	Each engine can be unit-tested/deployed/modified with defined interfaces and no cross-module data mutation.
NFR-09	Dashboards shall clearly show inventory state, recommendations, transfer status and required user actions.	Usability	A trained operator can identify current request/transfer state and next action within <=3 interactions.
NFR-10	Inventory and transfer operations shall be traceable through timestamped user, action, entity and state-change records.	Auditability	100% of inventory-changing and approval/status events generate an audit record.
NFR-11	Failed operations shall be recoverable without losing valid inventory state or creating duplicate transactions.	Recoverability	100% of defined failure/retry tests end in one valid consistent state.
NFR-12	The web application shall work on commonly used modern browsers and standard desktop/mobile screen sizes used by warehouse staff.	Compatibility	Pass functional UI tests on at least 2 modern browsers and responsive layouts for desktop + mobile viewport classes.

EVIDENCE 2 OF 7 — USE CASES / USER STORIES

Stage S1 — Functional Requirements Engineering | General, Theme-Independent

1. Use Case / User Story Table

One row per use case or user story. Cover normal, boundary, and exception flows.

ID	Actor	Description / User Story	Main Flow	Exception Flow
UC-01	Warehouse Staff / Manager	Monitor inventory and product status	System retrieves current stock, reservations, safety stock, DOI and demand indicators for selected warehouse/product.	If data is stale, show last-update time and flag the record for refresh; do not silently use stale values.
UC-02	Operations Manager	Detect stock risk and create redistribution request	Risk engine evaluates DOI, demand velocity and stock state; when redistribution is required, system creates a request.	If data is stale or demand changes before creation, re-evaluate before issuing the request.
UC-03	Operations Manager	Find and rank feasible source warehouses	System filters by stock surplus, safety stock, operational status and source demand risk, then calculates weighted scores.	If no single source is sufficient, invoke multi-source/partial fulfillment; if no safe source exists locally, consider regional/mother fallback.
UC-04	Operations Manager	Optimize source-to-destination route	For each selected source, system computes a route on the warehouse graph and returns distance/ETA.	If graph connectivity fails, return an alternate route or explicit no-route condition.
UC-05	Operations Manager	Review and approve/reject transfer	Manager reviews quantity, source score, route, ETA and source impact, then approves or rejects.	If source demand/stock changes before approval, invalidate stale recommendation and re-rank.
UC-06	System / Warehouse Staff	Reserve approved transfer stock	System atomically reserves approved quantity at the selected source(s).	If concurrent reservations consume the surplus, block/reduce reservation and re-evaluate instead of over-allocating.
UC-07	Logistics Staff	Accept assignment and update transfer status	Available logistics resource accepts the transfer and updates Assigned/Picked Up/In Transit/Delivered/Received.	If assignment fails or status stalls, reassign/escalate and notify the responsible manager.
UC-08	Warehouse Staff	Verify receipt and update inventory	Destination verifies received quantity; system finalizes source/destination balances and records the transaction.	If quantity differs, record discrepancy and require reconciliation before final closure.
UC-09	Operations Manager	Handle source-demand shock or regional shortage	System detects changed source demand or local shortage and re-evaluates pending transfers, including regional/mother warehouse fallback.	If continuing the transfer would breach source safety stock, revise quantity/source or split the request.
UC-10	Operations Manager / Management	Review dashboard, notifications and audit history	User sees role-based KPIs, active requests, transfers, alerts and complete transaction history.	If an audit/notification event is missing, flag it for reconciliation and retain the underlying transaction state.

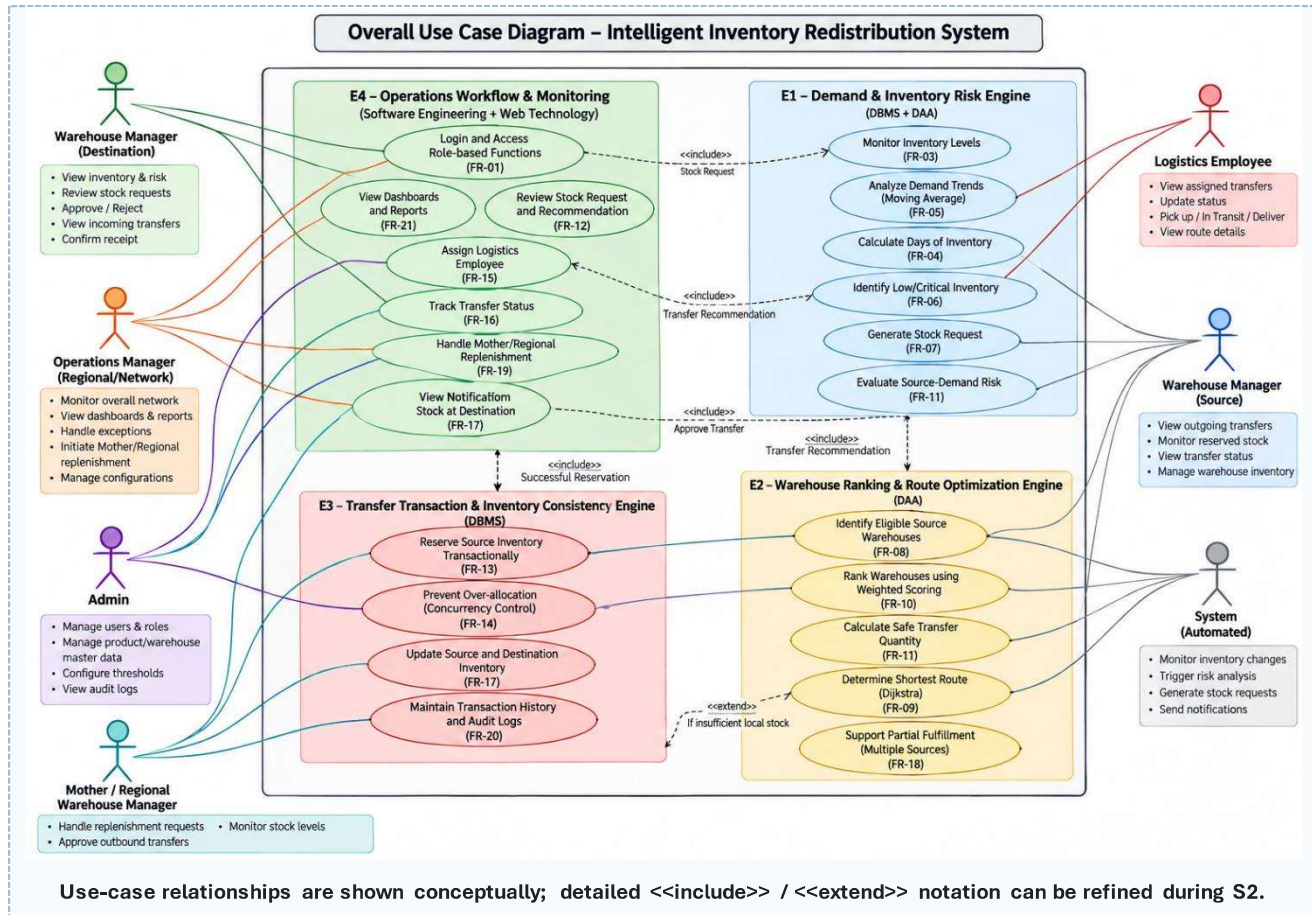
2. Use Case Diagram

Source-ranking rule used by E2: $\text{Score} = w_1 \cdot (1 - \text{distance_norm}) + w_2 \cdot \text{surplus_norm} + w_3 \cdot (1 - \text{source_risk_norm}) + w_4 \cdot (1 - \text{cost_norm}) + w_5 \cdot (1 - \text{ETA_norm})$, with $\sum w = 1$. Unsafe

candidates are filtered before scoring. Greedy allocation selects the highest-ranked feasible source(s); Dijkstra computes the shortest valid route on the warehouse graph.

Warehouse hierarchy rule: Local dark stores are evaluated first. A regional/mother warehouse is a fallback tier when local safe surplus cannot satisfy the request; long-distance transfer is accepted only when its total score/ETA/cost and source-safety constraints justify it.

Draw the complete use case diagram for the proposed system. Show all primary actors, system boundary, major use cases, and relevant <<include>> / <<extend>> relationships where applicable. The diagram must be consistent with the use cases or user stories listed above.



CHECK: Every actor and major use case shown in the diagram must correspond to the use case / user story table.

EVIDENCE 3 OF 7 — ACCEPTANCE CRITERIA SHEET
Stage S1 — Functional Requirements Engineering | General, Theme-Independent

1. Acceptance Criteria per Requirement

For every requirement, state the specific, testable condition(s) that define it as 'done'.

EVIDENCE 4 OF 7 — REQUIREMENT PRIORITY TABLE
Stage S1 — Functional Requirements Engineering | General, Theme-Independent

1. Priority Classification

Classify every requirement as Mandatory or Challenge/Stretch, with justification.

Requirement ID	Description	Priority	Justification
FR-01	The system shall authenticate authorized users and provide role-specific access to permitted functions.	Mandatory	Core acceptance requirement.
FR-02	The system shall maintain product, warehouse and inventory information, including available, reserved and safety-stock quantities.	Mandatory	Core acceptance requirement.
FR-03	The system shall monitor inventory levels across all participating warehouses.	Mandatory	Core acceptance requirement.
FR-04	The system shall calculate Economic Order Quantity (EOQ) for replenishment planning where the product has applicable demand/cost parameters.	Challenge	Core acceptance requirement.
FR-05	The system shall analyze historical sales and demand trends to derive demand velocity for products.	Mandatory	Core acceptance requirement.
FR-06	The system shall classify product inventory status as healthy, low or critical using configurable stock thresholds and demand-aware indicators.	Mandatory	Core acceptance requirement.
FR-07	The system shall automatically generate a stock redistribution request when Days of Inventory (DOI) falls below the configured trigger (baseline: DOI < 1), subject to validation of current stock and demand data.	Mandatory	Core acceptance requirement.
FR-08	The system shall compute a valid route between source and destination warehouses using a weighted warehouse graph and shortest-path algorithm where applicable.	Challenge	Core acceptance requirement.
FR-09	The system shall identify eligible source warehouses using transferable stock, safety-stock constraints, warehouse operational status and current source demand risk.	Mandatory	Core acceptance requirement.
FR-10	The system shall rank eligible source warehouses using a configurable weighted scoring mechanism based on distance, transferable surplus, source risk, transfer cost and ETA.	Mandatory	Core acceptance requirement.
FR-11	The system shall calculate a recommended transfer quantity that satisfies destination need without reducing the source below its safety-stock boundary.	Challenge	Core acceptance requirement.
FR-12	The system shall allow an authorized warehouse/operations manager to review, approve or reject a transfer request.	Mandatory	Core acceptance requirement.

Requirement ID	Description	Priority	Justification
FR-13	The system shall reserve approved source inventory transactionally before dispatch.	Mandatory	Core acceptance requirement.
FR-14	The system shall prevent over-allocation when multiple transfer requests for the same stock are processed concurrently.	Mandatory	Core acceptance requirement.
FR-15	The system shall automatically assign an available logistics employee/resource to an approved transfer.	Mandatory	Core acceptance requirement.
FR-16	The system shall track transfer status through Assigned → Picked Up → In Transit → Delivered → Received with timestamps.	Mandatory	Core acceptance requirement.
FR-17	The system shall update source and destination inventory after transfer receipt is confirmed and verified.	Mandatory	Core acceptance requirement.
FR-18	The system shall support partial/multi-source fulfillment when one eligible warehouse cannot safely satisfy the complete requested quantity.	Mandatory	Core acceptance requirement.
FR-19	The system shall support mother/regional warehouse replenishment as a fallback when suitable local stock is unavailable, while considering long-distance transfer cost, ETA and source safety.	Mandatory	Core acceptance requirement.
FR-20	The system shall maintain transfer history and an audit log for approvals, reservations, assignments, status changes and inventory changes.	Mandatory	Core acceptance requirement.
FR-21	The system shall provide role-based dashboards showing inventory levels, stock-risk states, requests, recommendations, transfers and operational status.	Mandatory	Core acceptance requirement.
FR-22	The system shall notify relevant users when a request is created, approved, rejected, assigned, delivered or completed.	Mandatory	Core acceptance requirement.

EVIDENCE 5 OF 7 — REQUIREMENTS TRACEABILITY MATRIX (RTM)
Stage S1 — Functional Requirements Engineering | General, Theme-Independent

1. Traceability Chain

Trace every requirement from its source need through to design, implementation and test.

Req ID	Source Need / (S0)	Owning Engine/Module	Design Ref / (S2–S4)	Test Ref / (S6)	Status
FR-01	secure role-based access	E4	S2 design TBD	S6 test TBD	Baseline
FR-02	reliable product/warehouse inventory visibility	E1	S2 design TBD	S6 test TBD	Baseline
FR-03	central monitoring of distributed stock	E1	S2 design TBD	S6 test TBD	Baseline
FR-04	replenishment planning using EOQ	E1	S2 design TBD	S6 test TBD	Baseline
FR-05	demand-aware stock decisions	E1	S2 design TBD	S6 test TBD	Baseline
FR-06	early stock-risk classification	E1	S2 design TBD	S6 test TBD	Baseline
FR-07	automatic redistribution trigger	E1	S2 design TBD	S6 test TBD	Baseline
FR-08	efficient source-destination routing	E2	S2 design TBD	S6 test TBD	Baseline
FR-09	safe source eligibility	E2	S2 design TBD	S6 test TBD	Baseline
FR-10	multi-factor source selection	E2	S2 design TBD	S6 test TBD	Baseline
FR-11	safe transfer quantity	E2	S2 design TBD	S6 test TBD	Baseline
FR-12	controlled manager approval	E4	S2 design TBD	S6 test TBD	Baseline
FR-13	transactional stock reservation	E3	S2 design TBD	S6 test TBD	Baseline
FR-14	concurrency-safe allocation	E3	S2 design TBD	S6 test TBD	Baseline
FR-15	automatic logistics assignment	E4	S2 design TBD	S6 test TBD	Baseline
FR-16	end-to-end transfer visibility	E4	S2 design TBD	S6 test TBD	Baseline
FR-17	accurate post-transfer inventory	E3	S2 design TBD	S6 test TBD	Baseline
FR-18	multi-source partial fulfillment	E2	S2 design TBD	S6 test TBD	Baseline
FR-19	regional/mother fallback replenishment	E2	S2 design TBD	S6 test TBD	Baseline
FR-20	traceability and auditability	E3	S2 design TBD	S6 test TBD	Baseline
FR-21	role-based operational visibility	E4	S2 design TBD	S6 test TBD	Baseline
FR-22	workflow notifications	E4	S2 design TBD	S6 test TBD	Baseline

ENGINE DECOMPOSITION TEMPLATE — STUDENT ENTRY
Stage S1 — Functional Requirements Engineering | Gate 1 Requirements Baseline

Purpose: derive independently ownable and testable technical engines from approved requirements before architecture or coding begins.

1. Requirement Clustering Sheet

Group related requirements by common technical responsibility. Do not cluster only by UI screen, frontend/backend layer, or convenience of work division.

	Cluster / Responsibility Area	Requirement IDs	Common Technical Responsibility	Cluster ID
C1	Inventory, demand & replenishment risk	FR-02, FR-03, FR-04, FR-05, FR-06, FR-07	Inventory state, demand trend, DOI, EOQ and risk classification	Demand & Inventory Risk Engine
C2	Source selection, transfer quantity & routing	FR-08, FR-09, FR-10, FR-11, FR-18, FR-19	Feasibility filtering, weighted ranking, multi-source allocation, fallback hierarchy and graph routing	Warehouse Ranking & Route Optimization Engine
C3	Transfer transaction & inventory consistency	FR-13, FR-14, FR-17, FR-20	Atomic reservation, concurrency control, inventory updates and audit trail	Transfer Transaction & Inventory Consistency Engine
C4	Operations workflow, logistics & monitoring	FR-01, FR-12, FR-15, FR-16, FR-21, FR-22	RBAC, approval workflow, logistics assignment, state machine, dashboards and notifications	Operations Workflow & Monitoring Engine

2. Engine Decomposition Matrix

Define each engine as a technical unit with a clear responsibility, inputs/outputs, owned state, mechanism, failure behaviour, measurable KPI and one primary student owner.

Engine ID	Engine Name	Requirement Cluster(s)	Primary Technical Responsibility	Inputs	Outputs	State / Data Owned	Core Mechanism / Algorithm	Dependencies / Interface	Failure Cases / Recovery	Primary KPI
E1	Demand & Inventory Risk Engine	C1 / FR-02, FR-03, FR-04, FR-05, FR-06, FR-07	Maintain inventory/demand state and compute replenishment and redistribution risk.	Product/warehouse use inventory; reservations; safety stock;	EOQ; health state; risk state; DOI; redistribution trigger; required quantity	Inventory snapshot; demand indicators; threshold configuration	Demand aggregation + moving/weighted demand velocity + DOI calculation + EOQ formula + configurable	DBMS inventory state; E3 reservations; E4 request workflow	Stale data; demand spike; missing history; duplicate trigger. Mark stale, re-evaluate,	Risk classification accuracy; risk evaluation <=2 s on pilot dataset; 100% bound

Engine ID	Engine Name	Requirement Cluster(s)	Primary Technical Responsibility	Inputs	Outputs	State / Data Owned	Core Mechanism / Algorithm	Dependencies / Interface	Failure Cases / Recovery	Primary KPI
				sales history; demand velocity; DOI inputs		n; risk state	threshold rules		suppress duplicate request and flag for review.	any cases deterministic
E2	Warehouse Ranking & Route Optimization Engine	C2 / FR-08, FR-09, FR-10, FR-11, FR-18, FR-19	Select safe source(s), calculate safe transfer quantities and compute efficient routes.	Destination request; warehouse graph; stock surplus; safety stock; source risk; distance; cost; ETA; warehouse tier	Eligible sources; weighted scores; selected source set; transfer quantities; route(s); explanation	Warehouse graph; candidate scores; allocation plan	Constraint filtering + weighted scoring + greedy multi-source allocation + Dijkstra shortest path	E1 risk; E3 reservation state; E4 request; warehouse master data	Nearest source unsafe; tie score; insufficient single source; disconnected route; long-distance regional fallback. Exclude unsafe candidates, split safely, use fallback tier or report no feasible plan.	Top-ranked feasible-source correctness; 0 safety-stock violations; route computation <=2 s
E3	Transfer Transaction & Inventory Consistency Engine	C3 / FR-13, FR-14, FR-17, FR-20	Preserve inventory correctness through reservation, concurrency control, receipt and audit.	Approved transfer; source balances; destination balances; received quantity; user/action context	Reservation result; updated balances; inventory transaction; audit record	Reservations; transfers; inventory transactions; audit records	DB transactions + row-level/optimistic locking + validation + idempotent state transitions + audit logging	E2 selected source; E4 approval/receipt; DBMS transaction layer	Duplicate reservation; concurrent request; partial receipt; retry. Reject duplicate allocation, rollback failed transaction, reconcile discrepancy and preserve valid	0 balance inconsistencies; 100% inventory-changing events audited

Engine ID	Engine Name	Requirement Cluster(s)	Primary Technical Responsibility	Inputs	Outputs	State / Data Owned	Core Mechanism / Algorithm	Dependencies / Interface	Failure Cases / Recovery	Primary KPI
									prior state.	
E4	Operations Workflow & Monitoring Engine	C4 / FR-01, FR-12, FR-15, FR-16, FR-21, FR-22	Coordinate role-based workflow from request through approval, logistics, status, dashboard and notifications.	Risk event ; recommendation; user action; logistics availability; transfer events	Request/approval/assignment/status state; role-specific dashboard data; notifications	Workflow state; roles/permissions; transfer timeline; notification events	State-machine workflow + RBAC + event handling + dashboard aggregation + notification rules	E1 risk event; E2 recommendation; E3 transaction state; WEB UI/API	Unauthorized action; no logistics resource; stale recommendation; notification retry. Deny action, reassign/escalate, trigger re-evaluation and use idempotent notifications.	>=95% workflow completion in pilot; 100% unauthorized actions rejected; status update latency <=2 s

ENGINE 01 — DESCRIPTION SHEET

S1 Engine Definition Template | Complete only for engines used by the team

Engine ID

E1

Engine Name

Demand & Inventory Risk Engine

Primary Student Owner — SRN

01FE24BCS254

Primary Student Owner — Name

Shrinidhi K

Requirement Cluster(s) / Requirement IDs

FR-02, FR-03, FR-04, FR-05, FR-06, FR-07

A. Engine Purpose and Description

In 3–5 sentences, state what this engine is responsible for, why it exists, and what technical problem it solves. Keep the description at S1 level; do not present detailed architecture or code.

A. Engine Purpose and Description

This engine maintains the current product/warehouse inventory and demand state and converts it into replenishment and redistribution risk signals. It calculates DOI, demand velocity and EOQ where applicable, then classifies inventory as healthy, low or critical using configurable rules. It triggers a redistribution request when the defined DOI/risk condition is reached while preventing duplicate or stale-data-driven requests.

B. Technical Responsibility and Boundary

Primary Responsibility	What This Engine Owns
Maintain inventory, demand and stock-risk state.	Inventory snapshot, reservations, safety stock, demand velocity, DOI, EOQ and risk state.

C. Inputs, Outputs and State

Inventory/product data; sales history; reservations; safety stock; current stock; trigger event.	EOQ; DOI; demand trend; healthy/low/critical state; redistribution trigger; required quantity.	Latest inventory state, demand indicators, threshold configuration and risk evaluation timestamp.
Inventory snapshot; reservations; safety stock; demand history/current velocity; stock-risk event.	Risk classification; required quantity; source-risk constraint.	Inventory state and latest risk evaluation timestamp.

D. Core Technical Mechanism

Algorithm / Mechanism / Rule	Why It Is Needed	Requirement(s) Supported
Demand aggregation + demand velocity + DOI + EOQ + configurable threshold/state rules.	A static “stock <= X” rule is insufficient for dynamic quick-commerce demand; the engine must account for rate of consumption and replenishment needs.	FR-02, FR-03, FR-04, FR-05, FR-06, FR-07

E. Dependencies, Failure Behaviour and KPI

Interface / Dependency	Possible Failure / Boundary Case	Expected Handling / Recovery	KPI / Measurement
DBMS inventory tables; sales history; E3 reservation state; E4 request workflow.	Stale/missing data; sudden demand spike; duplicate DOI trigger; insufficient history.	Show last-known timestamp, re-evaluate when data refreshes, suppress duplicate active requests and flag uncertain cases.	Risk classification accuracy >=95% on labelled pilot scenarios; evaluation <=2 s; 100% boundary cases deterministic.

ENGINE 02 — DESCRIPTION SHEET	
S1 Engine Definition Template Complete only for engines used by the team	
Engine ID	E2
Engine Name	Warehouse Ranking & Route Optimization Engine
Primary Student Owner — SRN	01FE24BCS183
Primary Student Owner — Name	Amruta J
Requirement Cluster(s) / Requirement IDs	FR-08, FR-09, FR-10, FR-11, FR-18, FR-19

A. Engine Purpose and Description

In 3–5 sentences, state what this engine is responsible for, why it exists, and what technical problem it solves. Keep the description at S1 level; do not present detailed architecture or code.

A. Engine Purpose and Description

This engine decides which warehouse or combination of warehouses should supply a redistribution request and how the transfer should be routed. It first filters unsafe candidates using stock surplus, safety stock, warehouse status and source-demand risk. It then applies a configurable weighted score and greedy multi-source allocation; Dijkstra is used for shortest-path routing on the warehouse graph. A mother/regional warehouse is treated as a fallback tier when local sources cannot safely satisfy the request.

B. Technical Responsibility and Boundary

Primary Responsibility	What This Engine Owns
Filter and rank feasible source warehouses, calculate safe transfer quantities and compute routes.	Candidate filtering, weighted score, multi-source allocation, selected source set, transfer quantities, route and explanation.

C. Inputs, Outputs and State

Destination request; warehouse graph; stock surplus; safety stock; source risk; distance; cost; ETA; warehouse tier.	Eligible sources; score breakdown; selected source(s); transfer quantities; route(s); fallback decision.	Warehouse graph, candidate scores and allocation plan.
Destination request; warehouse graph; source surplus; source-demand risk; distance/ETA/cost.	Ranked candidates; selected source; score breakdown; route.	Warehouse graph and candidate ranking state.

D. Core Technical Mechanism

Algorithm / Mechanism / Rule	Why It Is Needed	Requirement(s) Supported
Constraint filtering + weighted scoring + greedy multi-source allocation + Dijkstra shortest path.	The nearest warehouse is not always the best source; the system must balance proximity with surplus, source risk, cost, ETA and long-distance fallback options.	FR-08, FR-09, FR-10, FR-11, FR-18, FR-19

E. Dependencies, Failure Behaviour and KPI

Interface / Dependency	Possible Failure / Boundary Case	Expected Handling / Recovery	KPI / Measurement
E1 risk; E3 reservation state; E4 request; warehouse master/graph data.	Nearest source unsafe; tie score; no single source sufficient; disconnected route; local stock unavailable.	Exclude unsafe candidates, split across safe sources, deterministic tie-break, use regional/mother fallback or return no feasible plan.	0 safety-stock violations; >=95% correct top-ranked source on test set; route <=2 s.

ENGINE 03 — DESCRIPTION SHEET <i>S1 Engine Definition Template Complete only for engines used by the team</i>	
Engine ID	E3
Engine Name	Transfer Transaction & Inventory Consistency Engine
Primary Student Owner — SRN	01FE24BCS110
Primary Student Owner — Name	Kiran M
Requirement Cluster(s) / Requirement IDs	FR-13, FR-14, FR-17, FR-20

A. Engine Purpose and Description

In 3–5 sentences, state what this engine is responsible for, why it exists, and what technical problem it solves. Keep the description at S1 level; do not present detailed architecture or code.

A. Engine Purpose and Description

This engine guarantees inventory correctness throughout reservation, dispatch, receipt and reconciliation. It uses transactional updates and concurrency control so approved stock cannot be allocated twice. It updates source and destination balances exactly once after verified receipt and records every inventory-changing event in an audit trail. It is the main DBMS/SE boundary for consistency and recovery.

B. Technical Responsibility and Boundary

Primary Responsibility	What This Engine Owns	What This Engine Does NOT Own
Maintain consistent inventory and traceable transfer transactions.	Reservations, concurrency control, transfer lifecycle state, inventory transactions and audit events.	It does not choose the source, calculate ranking scores, or perform physical transportation.

C. Inputs, Outputs and State

Approved transfer; source/destination balances; received quantity; user/action context; concurrent requests.	Reservation result; updated balances; inventory transaction; audit record; recovery outcome.	Reservation state, transfer state, inventory transaction and audit state.
Approved transfer; source inventory; received quantity; user/action context.	Reservation result; updated balances; inventory transaction; audit record.	Reservation status, transfer state, inventory transaction and audit state.

D. Core Technical Mechanism

Algorithm / Mechanism / Rule	Why It Is Needed	Requirement(s) Supported
DB transactions + locking/concurrency control + validation + idempotent state transitions + audit logging.	Without atomicity and concurrency control, simultaneous requests can allocate the same stock or leave source/destination balances inconsistent.	FR-13, FR-14, FR-17, FR-20; NFR-02, NFR-04, NFR-05, NFR-10, NFR-11

E. Dependencies, Failure Behaviour and KPI

Interface / Dependency	Possible Failure / Boundary Case	Expected Handling / Recovery	KPI / Measurement
E2 source/allocation plan; E4 approval/receipt; DBMS transaction layer.	Duplicate reservation; concurrent request; partial receipt; retry; process failure.	Reject duplicate allocation, rollback failed transaction, reconcile receipt discrepancy, keep idempotent state and log exception.	0 balance inconsistencies; 100% inventory-changing events audited; 100% concurrency tests preserve valid state.

ENGINE 04 — DESCRIPTION SHEET <i>S1 Engine Definition Template Complete only for engines used by the team</i>	
Engine ID	E4
Engine Name	Operations Workflow & Monitoring Engine
Primary Student Owner — SRN	01FE24BCS185
Primary Student Owner — Name	Rajat P
Requirement Cluster(s) / Requirement IDs	FR-01, FR-12, FR-15, FR-16, FR-21, FR-22

A. Engine Purpose and Description

In 3–5 sentences, state what this engine is responsible for, why it exists, and what technical problem it solves. Keep the description at S1 level; do not present detailed architecture or code.

A. Engine Purpose and Description

This engine coordinates the operational workflow from redistribution request through approval, logistics assignment, movement tracking, dashboard monitoring and notifications. It enforces role-based actions and a valid transfer state machine while keeping users informed of state changes. It does not perform the ranking or inventory transaction itself; it orchestrates the other engines through defined interfaces.

B. Technical Responsibility and Boundary

Primary Responsibility	What This Engine Owns	What This Engine Does NOT Own
Coordinate role-based request, approval, logistics, status, dashboard and notification workflow.	User roles/permissions, workflow states, assignment status, dashboard aggregation and notification events.	It does not calculate source ranking, own inventory balances or physically move goods.

C. Inputs, Outputs and State

Risk/request event; ranked recommendation; manager action; logistics availability; transfer events.	Request/approval/assignment/status transitions; dashboard data; notifications; escalation events.	Workflow state, roles/permissions, transfer timeline and notification event state.
Risk event; ranked recommendation; manager decision; logistics availability/status.	Request/approval/assignment/status transitions and exception events.	Workflow state, role permissions and transfer status timeline.

D. Core Technical Mechanism

Algorithm / Mechanism / Rule	Why It Is Needed	Requirement(s) Supported
State-machine workflow + RBAC + event handling + dashboard aggregation + idempotent notification rules.	Operational actions must occur in a controlled order and only authorized roles should perform them; users need immediate state visibility.	FR-01, FR-12, FR-15, FR-16, FR-21, FR-22; NFR-03, NFR-06, NFR-08, NFR-09, NFR-12

E. Dependencies, Failure Behaviour and KPI

Interface / Dependency	Possible Failure / Boundary Case	Expected Handling / Recovery	KPI / Measurement
E1 risk; E2 recommendation; E3 transaction state; web UI/API; notification service.	Unauthorized action; no logistics resource; stale recommendation; delayed status; notification retry.	Deny unauthorized action, reassign/escalate, invalidate stale recommendation, and use idempotent notification handling.	>=95% workflow completion in pilot; 100% unauthorized actions rejected; status updates <=2 s.

